Module-Wrestler Online Character Dictionary

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Abstract

We are the most modern iteration of a long-established translation dictionary relating the language of characters with the language of finite complex group modules.

1 The Translation Dictionary

There are categories **FinGroup** of finite groups, **FinVect** of finite-dimensional vector spaces over $\mathbb{C}$, and for each $G : \mathbf{FinGroup}$, a category $\mathbf{FinVect}^G$ of finite-dimensional complex G-modules. If $V : \mathbf{FinVect}^G$, the G-action will be denoted $G \xrightarrow{\rho_V} \mathrm{GL}(V)$ or $g \cdot x := \rho_V(g)(x)$. If it is clear, we may denote $\rho_V(g)$ by $g \cdot -$. The set of conjugacy classes will be denoted $\mathcal{CC}(G)$. For $x \in G$, its conjugacy class is denoted $\mathcal{CL}(x)$.

1.1 Definitions in the Language of Characters

Throughout, $G : \mathbf{FinGroup}$ is a group and $V : \mathbf{FinVect}^G$ is a finite-dimensional complex G-module. (Any other type data will be specified.)

(D1) **Character** of V: The function $\chi_V := \mathrm{tr} \circ \rho_V : G \to \mathbb{C}$. It is a class function (D3), so it is occasionally factored through the quotient $G \twoheadrightarrow \mathcal{CC}(G)$, where it is regarded as a map $\mathcal{CC}(G) \to \mathbb{C}$.

(D2) **Character table** of G: A listing of the characters χ_I corresponding to the finite-dimensional complex irreducible modules I of G. As characters are class functions (D3), the value on each *conjugacy class* is listed. As the irreducible modules biject with the conjugacy classes, this is a square matrix. For example, behold the character table of S_2 (it's not interesting):

Character	$(*)(*)$	$(**)$
χ_{trivial}	1	1
χ_{sign}	1	-1

Beware – although the row for χ_I encodes the character χ_I, it is missing information on the size of the conjugacy class. This information is critical to computing the inner product of two characters (D4); generically, the inner product **does not** agree with the "dot product".

(D3) **Class function**: A function $\phi : G \to A$ which is constant on conjugacy classes of G. A is typically $\mathbb{C}$.

(D4) **Inner product** on $\mathbb{C}^G$: The vector space of functions $G \to \mathbb{C}$ carries Hermitian inner product $\langle -, - \rangle$ given by

$$\langle \phi, \psi \rangle := \frac{1}{|G|} \sum_{g \in G} \phi(g) \cdot \overline{\psi(g)}$$

or, for characters $(U, V : \mathbf{FinVect}^G)$:

$$\langle \chi_U, \chi_V \rangle := \frac{1}{|G|} \sum_{g \in G} \chi_U(g) \cdot \chi_V(g^{-1})$$

because of (G1).

1.2 Grammatical Rules of the Language of Characters

I mean in this section to list some key properties that characters always satisfy. If you were to communicate using characters, these are some rules that "always appear" in your sentences! Throughout, $G : \mathbf{FinGroup}$ and $U, V : \mathbf{FinVect}^G$ unless otherwise specified.

(G1) **Inverses**: For $g \in G$, $\chi_V(g^{-1}) = \overline{\chi_V(g)}$.

(G2) **Frobenius reciprocity**: If $H \leq G$, $U : \mathbf{FinVect}^H$ and $V : \mathbf{FinVect}^G$, then

$$\langle \mathrm{Ind}_H^G U, V \rangle = \langle U, \mathrm{Res}_H^G V \rangle.$$

See (T1).

(G3) **Orthonormality of irreducibles**:

$$U, V \text{ both irreducible} \implies \langle \chi_U, \chi_V \rangle = \mathbf{1}_{U \simeq V} \quad \text{(indicator function)}.$$

1.3 Translating Between Characters and Representations

Throughout, $G : \mathbf{FinGroup}$ and $U, V : \mathbf{FinVect}^G$.

(T1) **Adjunction of induction to restriction**: For a subgroup $H \leq G$, the adjunction of functors $\mathrm{Ind}_H^G \dashv \mathrm{Res}_H^G$ appears as **Frobenius reciprocity**:

$$\mathbf{FinVect}^G(\mathrm{Ind}_H^G U, V) \simeq_{\mathrm{nat}.U,V} \mathbf{FinVect}^H(U, \mathrm{Res}_H^G V) \quad \rightsquigarrow \quad \langle \mathrm{Ind}_H^G U, V \rangle = \langle U, \mathrm{Res}_H^G V \rangle.$$

The inner product relation is a decategorification of the (categorical) adjunction.

(T2) **Dimension/Degree** of V:

$$\dim(V) = \chi_V(1).$$

(T3) **Dimension** of hom-space $\mathbf{FinVect}^G(U, V)$ (this space goes by the notation $\mathrm{Hom}_G(U, V)$ in the course):

$$\dim \mathbf{FinVect}^G(U, V) = \langle \chi_U, \chi_V \rangle.$$

(T4) **Equivalence of irreducible modules**: In the context where U, V are irreducible,

$$U \simeq V \iff \langle \chi_U, \chi_V \rangle = 1$$

and furthermore, in the same context,

$$U \not\simeq V \iff \langle \chi_U, \chi_V \rangle = 0.$$

See (G3) for a more concise statement.

(T5) **Frobenius reciprocity**: See (T1).

(T6) **Irreducible factors**: For U irreducible,

$$\text{a factor of } U \text{ appears in } V \iff \langle \chi_U, \chi_V \rangle \neq 0.$$

(A factor "appearing" means that $V \simeq U \oplus (\text{stuff})$.)

(T7) **Irreducibility**: See also (G3).

$$V \text{ is irreducible} \iff \langle \chi_V, \chi_V \rangle = 1.$$

2 The Story of Characters, Told in Broad Strokes

If you want, this section charts the "etymology" of each word in the dictionary!

The slogan is that **characters decategorify modules**, in analogy to how **dimension decategorifies vector spaces**.

$$\begin{array}{ccc} \text{Categorified} & \xrightarrow{\text{character}} & \text{Decategorified} \\ \text{Module } U & \longmapsto & \text{List of numbers } \chi_U \in \mathbb{C}^G \\ \downarrow \text{morphism space} & & \downarrow \text{its dimension} \\ \text{Module } V & \longmapsto & \text{List of numbers } \chi_V \in \mathbb{C}^G \end{array} \qquad \begin{array}{c} \text{What is lost?} \\ \text{Algebraic data (action } G \to \mathrm{GL}(U)) \\ \downarrow \text{transformation of maps} \\ \text{Algebraic data: (action } G \to \mathrm{GL}(V)) \end{array}$$

To exemplify the analogy:

$$\begin{array}{ccc} \text{Categorified} & \xrightarrow{\text{dimension}} & \text{Decategorified} \\ \text{Vect. Sp. } U & \longmapsto & \text{Number } \dim U \in \mathbb{Z}_{\geq 0} \\ \downarrow \text{morphism space} & & \downarrow \text{rank-nullity statements} \\ \text{Vect. Sp. } V & \longmapsto & \text{Number } \dim V \in \mathbb{Z}_{\geq 0} \end{array} \qquad \begin{array}{c} \text{What is lost?} \\ \text{Algebraic data (vectors)} \\ \downarrow \text{transformation of algebraic data} \\ \text{Algebraic data (vectors)} \end{array}$$

In both situations, **the decategorification classifies the categorification up to isomorphism** – that is, two (finite-dimensional) vector spaces are isomorphic just when they have the same dimension, and two (finite-dimensional, complex) G-modules are isomorphic just when they have the same character. At the level of objects, the decategorification retains all necessary information (as one would hope of anything being granted the monacre of "decategorification"). However, passing to this "more computational invariant" has the deficiency of losing algebraic data. On decategorifying **FinVect**, we lose large amounts of data on how linear maps put together vector spaces. On decategorifying $\mathbf{FinVect}^G$, we lose large amounts of data on the actual action of G, and we lose information on how these actions intertwine.

We now tell a fiction on the story of character theory, all in broad strokes. Once upon a time, vector spaces were decategorified by their dimension, and everyone lived happily ever after. Well, they lived up until algebraists decided to represent symmetries on vector spaces, moving us from the land **FinVect** to the various lands $\mathbf{FinVect}^G$ (for $G : \mathbf{FinGroup}$). In early and tumultuous days, many discoveries were made paralleling the story for vector spaces:

	Vector spaces	Modules
can be broken up into	one-dimensional subspaces,	irreducible submodules;
the maps between which are	effectively scalars,	effectively scalars;
and which form all objects by	direct sum,	direct sum.

After these initial algebraic gains, however, lingered important questions – "How many irreducible modules are there?", or "Where can we find these modules?", for example. To gain further ground, the category $\mathbf{FinVect}^G$ needed to be quite seriously tamed. After dusting off the history books, the algebraists attempted to decategorify $\mathbf{FinVect}^G$ by mirroring the dimension decategorification of **FinVect**. Fortunately, in the context of a group representation, the dimension appears naturally:

$$G \xrightarrow{\rho_V} \mathrm{GL}(V) \quad \rightsquigarrow \quad \rho_V(1_G) = 1_V = \underbrace{\begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}}_{n \times n} \quad \rightsquigarrow \quad \dim(V) = n = \mathrm{tr}(\rho_V(1_G)).$$

This is a natural choice because the representing linear transformations $\rho_V(g), g \in G$ are the only extra data present within objects of $\mathbf{FinVect}^G$, and the only such transformation guaranteed to exist is $\rho_V(1_G) = 1_V$. The generalisation easily revealed itself – the **character** of V is the function $\chi_V : G \to \mathbb{C}$ given by $\chi_V := \mathrm{tr} \circ \rho_V$. The miracle is that this successfully decategorified away enough complexity to make tractable progress in understanding the category $\mathbf{FinVect}^G$.

Perhaps the preceding text is best understood as a prologue to our story – "Chapter 0", if you will. In chapter one, we begin to understand the hom-spaces $\mathbf{FinVect}^G(U, V)$ for $U, V : \mathbf{FinVect}^G$. Every part of our story takes place over the category of vector spaces, so it is not a surprise to find that $\mathbf{FinVect}^G(U, V)$ is a vector space. Recording its dimension would be a first feat, and no easy one at that. In perhaps a stroke of luck, or perhaps by arcane insight dug up from historical texts[1], one can find some inspiration in a superficial look at $\mathbf{FinVect}^G(-,-)$:

$$\begin{array}{lll} \text{Categorified} & \mathbf{FinVect}^G \times \mathbf{FinVect}^G \longrightarrow \mathbf{FinVect} & \\ & (U, V) \longmapsto \mathbf{FinVect}^G(U, V), & \\ \text{character} \downarrow & & \\ \text{Decategorified} & \mathbb{C}^G \times \mathbb{C}^G \dashrightarrow \mathbb{C} & \\ & (\chi_U, \chi_V) \longmapsto ?. & \end{array}$$

The superficial symmetry is that $\mathbf{FinVect}^G(-,-)$ appears to decategorify into something resembling an inner product. There is a natural choice of inner product on the set of functions $G \to \mathbb{C}$, given by

$$\begin{aligned} \mathbb{C}^G \times \mathbb{C}^G &\xrightarrow{\langle -,- \rangle} \mathbb{C} \\ (\phi, \psi) &\longmapsto \frac{1}{|G|} \sum_{g \in G} \phi(g) \cdot \overline{\psi(g)}. \end{aligned}$$

The division by $|G|$ may be seen as a normalisation factor; this had proven useful to the algebraists prior. The miracle of chapter one is that *this inner product idea worked* and gave rise to a rich theory, eventually culminating in the dimension formula

$$\dim \mathbf{FinVect}^G(U, V) = \langle \chi_U, \chi_V \rangle$$

and answering some of the lingering questions prior. How these results follow is not immediate, and will take us a couple more chapters to describe. We begin right now with chapter two.

FIXME: do it

3 Proofs

We collect together proofs of all statements in the translation dictionary and in the "story told in broad strokes" (sections 1 and 2).

4 Some Character Tables

[1] The *Module-Wrestler Online Character Dictionary* keeps no good record of history.